Cambridge (CIE) A Level Maths: Pure 3

Your notes

Logarithmic & Exponential Functions

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- Exponential Functions
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Your notes

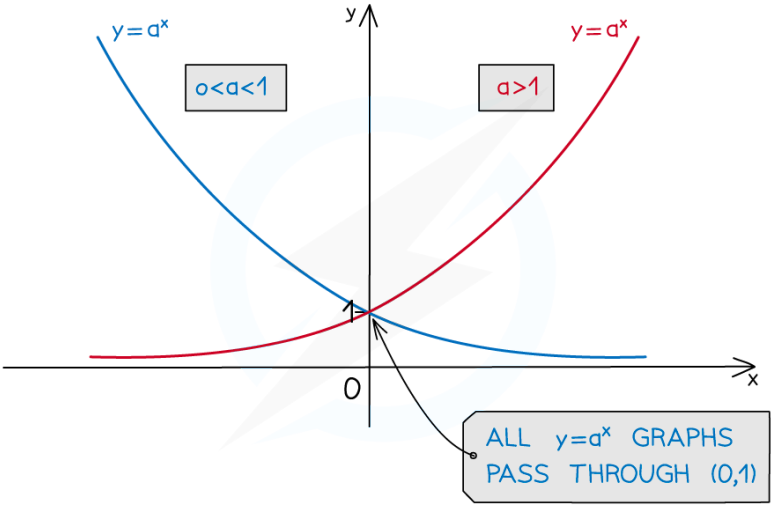
Exponential functions

What are exponential functions?

- Exponential functions of the form **y = a^x** with **a > 0** are considered at A level

What are exponential graphs?

- All graphs of the form **y = a^x** will pass through (0, 1) because **a^0 = 1**
- The **x**-axis is an **asymptote**



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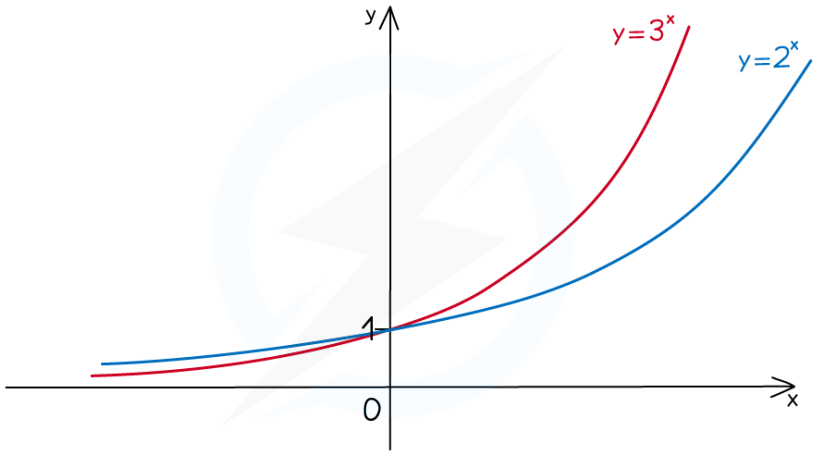
What is exponential growth?

- Where **x < 0** the higher value of **a** is the "lower" graph
- Where **x > 0** the higher value of **a** is the "higher" graph
- **a > 1** is **exponential growth** (see Exponential Growth & Decay)



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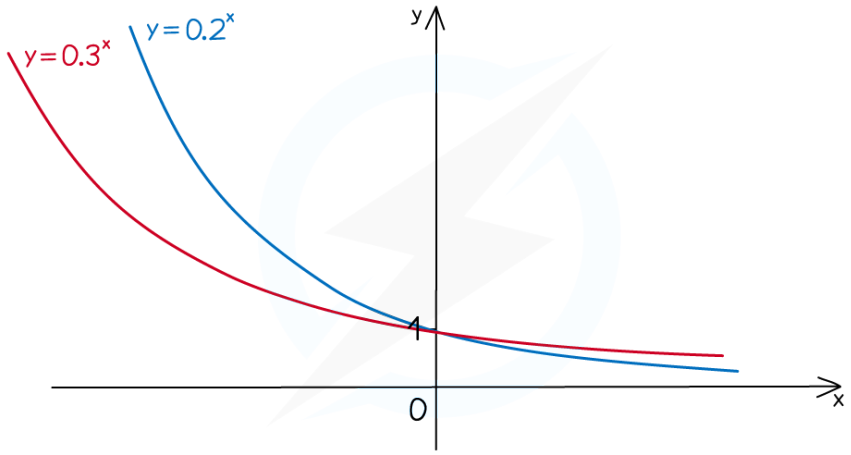
What is exponential decay?

- Where **x < 0** the higher value of **a** is the "lower" graph
- Where **x > 0** the higher value of **a** is the "higher" graph
- **0 < a < 1** is **exponential decay**



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- You may like to think about why **a = 1** is **not** considered... If a = 1, $y = 1^x = 1$ for all values of x

Worked Example

Your notes

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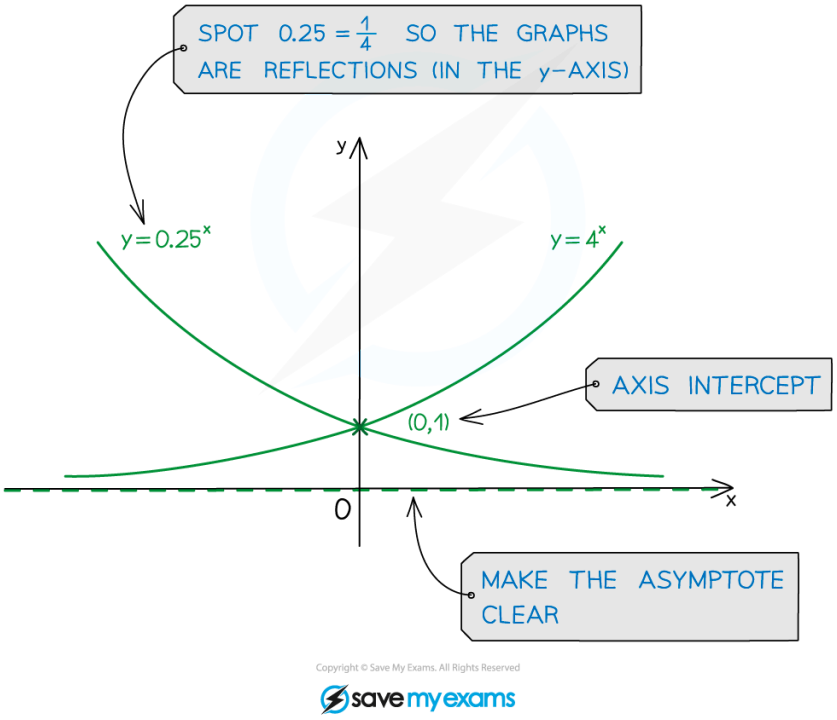
On the same diagram sketch the graphs of

$y = 4^x$ and $y = 0.25^x$.

Mark any points where the graphs cross the coordinate axes and make any asymptotes clear.



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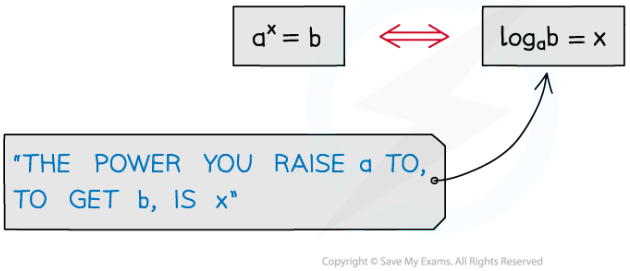
Logarithmic functions

What are logarithmic functions?



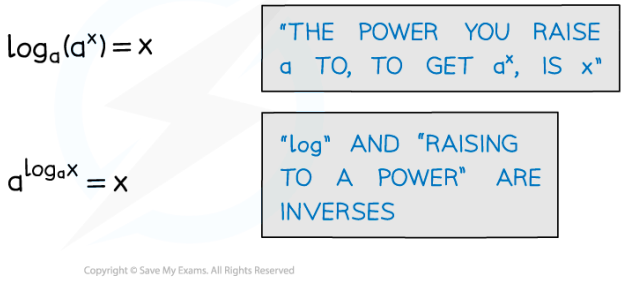
- **a = b^x** and **log $_b$ a = x** are equivalent statements
- **a > 0**
- **b** is called the base
- Every time you write a logarithm statement say to yourself what it means
  - **$\log_3 81 = 4$**

    "the power you raise 3 to, to get 81, is 4"
  - **$\log_p q = r$**

    "the power you raise **p** to, to get **q**, is **r**"

How are logarithms and powers inverses of each other?

- A logarithm is the inverse of raising to a power so we can use rules to simplify logarithmic functions



How do I use the inverse property of logs and powers?

Your notes

e.g. WRITE $\dfrac{2^{\log_3 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}}$ IN THE FORM $\dfrac{a}{b}$

WHERE a AND b ARE INTEGERS TO BE FOUND.

"THE POWER YOU RAISE 3 TO, TO GET 81, IS 4"

$2^{\log_3 81} = 2^4 = 16$

"THE POWER YOU RAISE 10 TO, TO GET 1000, IS 3"

$3^{\log_{10} 1000} = 3^3 = 27$

$\log_5 5 = \log_5 5^1 = 1$

$5^{\log_5 5} = 5^1 = 5$

$$\therefore \quad \frac{2^{\log_3 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}} = \frac{16 + 27}{5}$$

$$= \frac{43}{5}$$

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- Recognising the rules of logarithms allows expressions to be simplified

e.g. WITHOUT USING A CALCULATOR,
EVALUATE $\log_2 64\sqrt{2}$

RECOGNISE 64 AND $\sqrt{2}$ ARE POWERS OF 2

"THE POWER YOU RAISE 2 TO, TO GET $64\sqrt{2}$, IS x"

LET $x = \log_2 64\sqrt{2}$

$2^x = 64\sqrt{2}$

REWRITE THE STATEMENT

$2^x = 2^6 \times 2^{\frac{1}{2}}$

$2^x = 2^{6+\frac{1}{2}}$

USING LAWS OF INDICES

$2^x = 2^{\frac{13}{2}}$

$x = \dfrac{13}{2}$

$\therefore \log_2 64\sqrt{2} = \dfrac{13}{2}$

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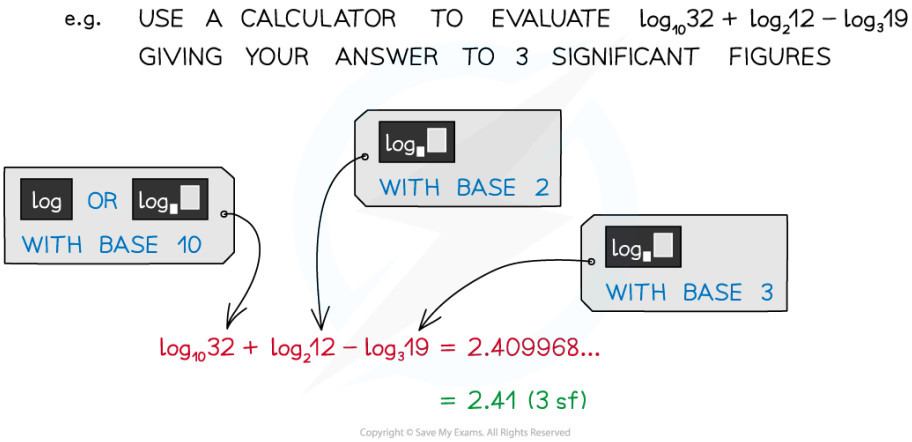
- Recognition of common powers helps in simple cases
  - Powers of 2: $2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, \ldots$
  - Powers of 3: $3^0 = 1, 3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, \ldots$
  - The first few powers of 4, 5 and 10 should also be familiar

  For more awkward cases a calculator is needed



- Calculators can have, possibly, three different logarithm buttons

- This button allows you to type in any number for the base

- Natural logarithms (see "e")

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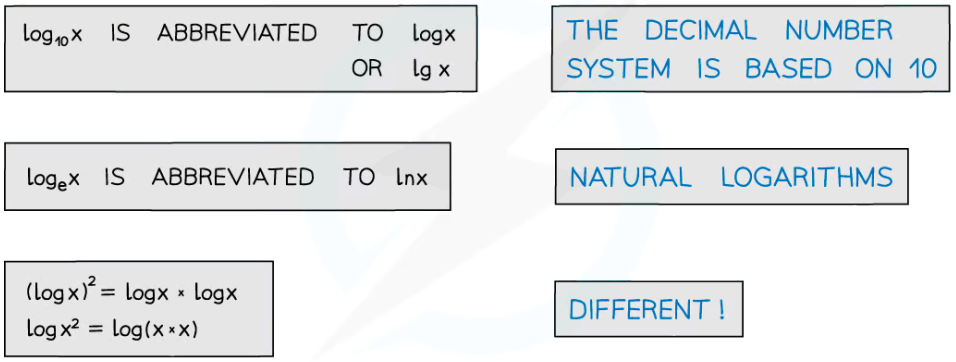
- Shortcut for base 10 although SHIFT button needed
- Before calculators, logarithmic values had to be looked up in printed tables

What notation is used with logarithms?



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- 10 is a common base
  - **$\log_{10} x$** is abbreviated to **$\log x$** or **$\lg x$**
- The value **e** is another common base
  - **$\log_e x$** is abbreviated to **$\ln x$**
- **$(\log x)^2 \neq \log x^2$**

Worked Example

Your notes

Solve the equation $2^{2x+3} - 41\left(2^{x}\right) + 5 = 0$,

giving your answers correct to 3 significant figures.

$2^{2x+3} - 41(2^x) + 5 = 0$

HIDDEN QUADRATIC

$2^3 \times 2^{2x} - 41(2^x) + 5 = 0$

$8(2^x)^2 - 41(2^x) + 5 = 0$

REWRITE

LET $y = 2^x$: $8y^2 - 41y + 5 = 0$

SUBSTITUTION NOT ESSENTIAL

$(8y-1)(y-5) = 0$

USE CALCULATOR TO SOLVE

$y = \frac{1}{8}$ $\quad$ $y = 5$

$2^x = \frac{1}{8}$ $\quad$ $2^x = 5$

SPOT THIS ?!

$x = \log_2 \frac{1}{8}$ $\quad$ $x = \log_2 5$

USE LOGARITHMS

$x = -3$ $\quad$ $x = 2.32192...$

$x = 2.32$ (3sf)

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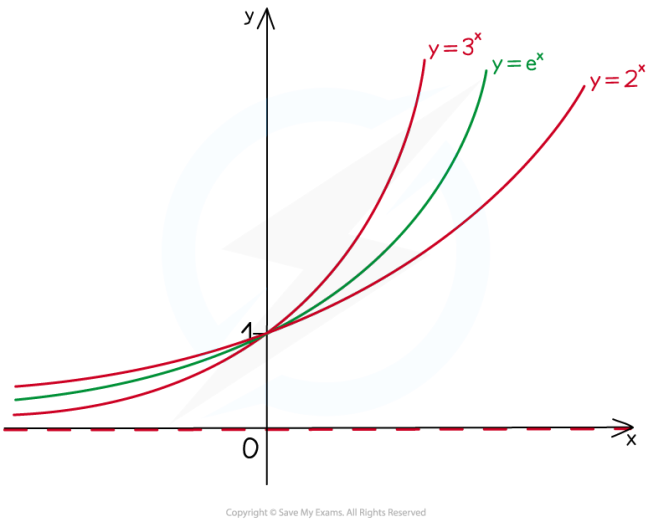
"e"

What is the number "e" in maths?

- **e** is an irrational number, sometimes called Euler's number
  - **e ≈ 2.718**

How do I sketch y = e^x?

- The exponential function is **y = e^x**
  - As with other exponential graphs **y = e^x**
    - passes through (0, 1)
    - has the x-axis as an asymptote

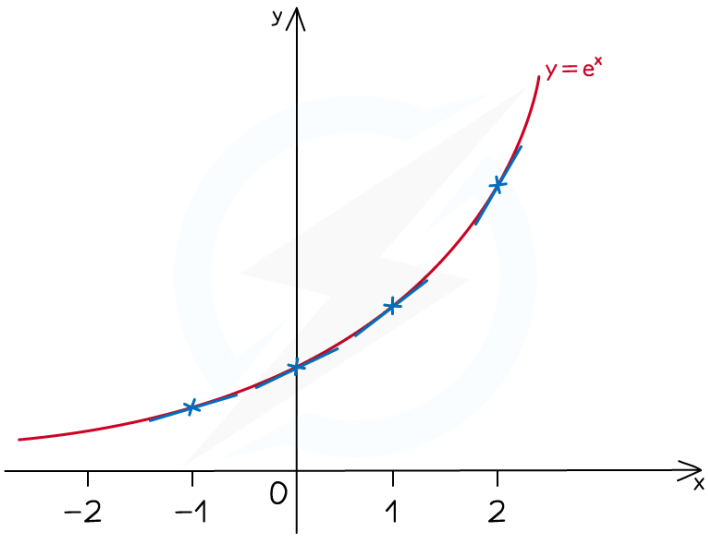


What is the special property of "e"?

- **y = e^x** has the particular property that

$$\frac{dy}{dx} = e^x$$

- ie for *every* real number **x**, the gradient of **y = e^x** is also equal to **e^x** (see Derivatives of Exponential Functions)



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x	y	dy / dx
−1	0.3678...	0.3678...
0	1	1
1	2.7182...	2.7182...
2	7.3890...	7.3890...

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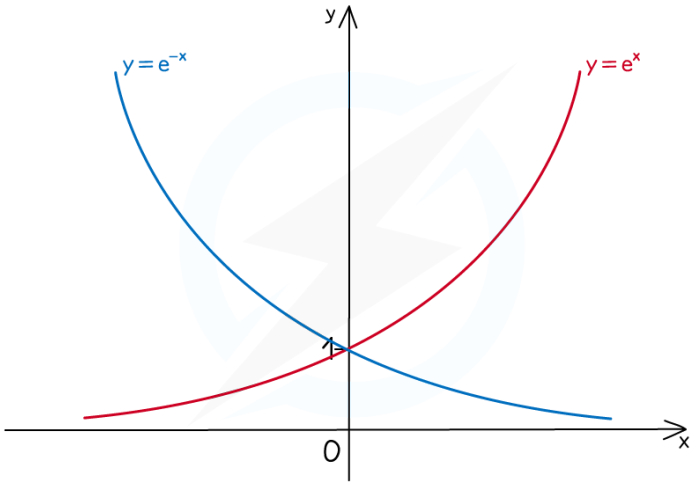
How do I sketch $y = e^{-x}$?

- $y = e^{-x}$ is a reflection in the **y**-axis of $y = e^{x}$
- They are of the form **y = f(x)** and **y = f(−x)** (see Transformations of Functions - Reflections)

Your notes



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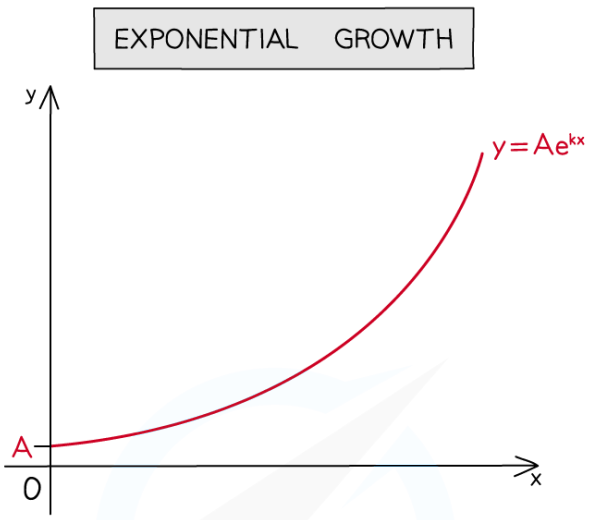
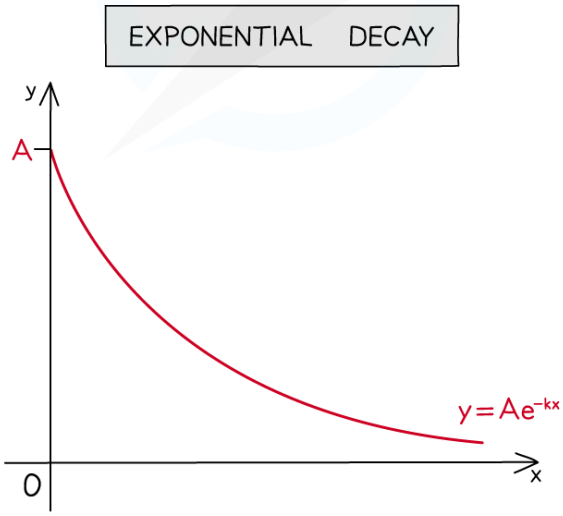
Your notes

What is exponential growth and decay?

Your notes





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- **$y = Ae^{kx}$ ($k > 0$)** is exponential growth
- **$y = Ae^{-kx}$ ($k > 0$)** is exponential decay
- **A** is the initial value
- **k** is a (usually positive) constant
- "–" is used in the equation making clear whether it is growth or decay

Worked Example

Your notes

(a) On the same diagram sketch the graphs of

(i) $y = e^{x}$

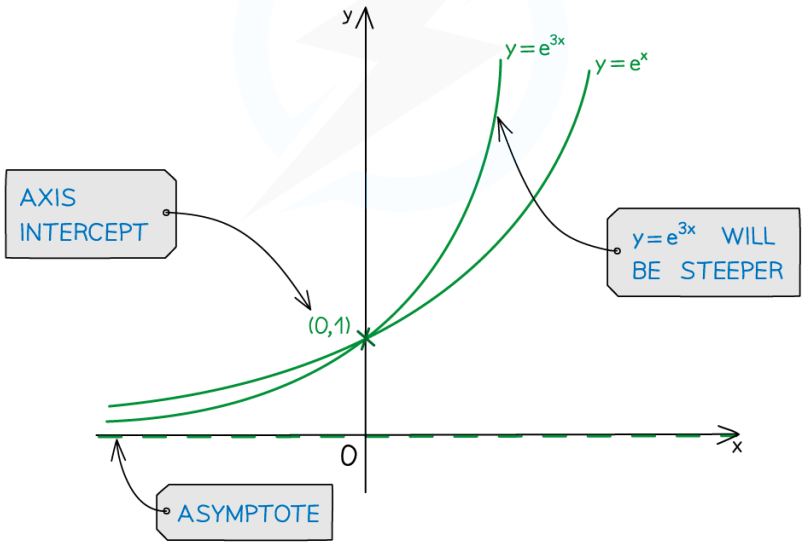
(ii) $y = e^{3x}$

(b) On the same diagram sketch the graphs of

(i) $y = e^{-x}$

(ii) $y = e^{-7x}$

a) (i) & (ii)



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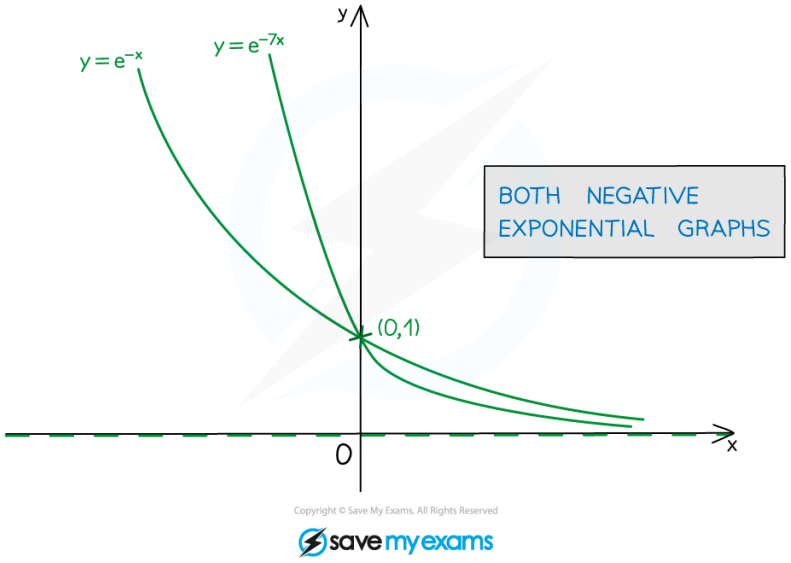
b) (i) & (ii)



Your notes

Your notes

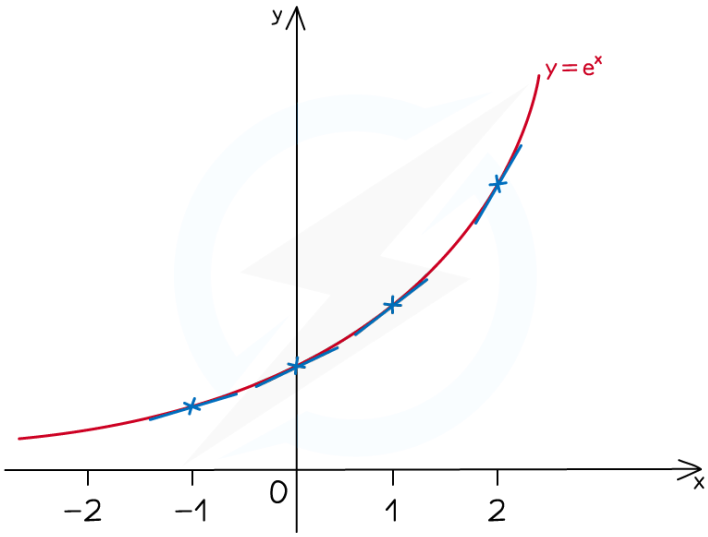
Derivatives of exponential functions

What is the derivative of e^x?

- **y = e^x** has the particular property that

$$\frac{dy}{dx} = e^x$$

- i.e. for *every* real number **x**, the gradient of **y = e^x** is also equal to **e^x** (see Derivatives of Exponential Functions)



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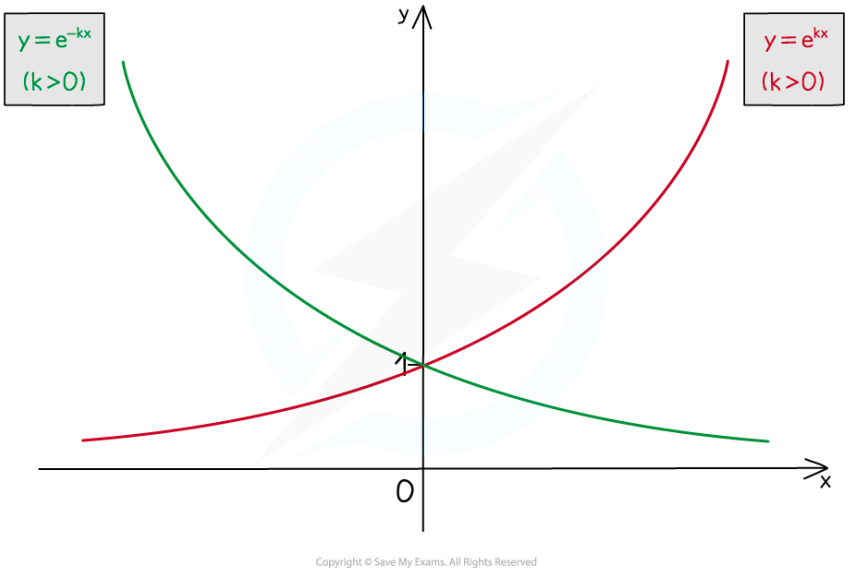
- **e ≈ 2.718** (see "e")

- Recall that the derivative is the gradient function for a curve (see First Principles Differentiation)

Your notes

What is the derivative of y = e^{kx} and y=e^{-kx}?



- The derivative of **y = e^{kx}** is
  - $$\frac{dy}{dx} = ke^{kx}$$
- The derivative of **y = e^{-kx}** is
  - $$\frac{dy}{dx} = -ke^{-kx}$$

Examiner Tips and Tricks

- Remember that (like **π**) **e** is a **number**.
- Exam questions can ask for answers to be given as exact values in terms of **e** (see the Worked Example below).

Worked Example

Your notes

Find, in terms of e, the gradient at $x = 3$ for each of the following functions ...

(a) $y = e^x$

(b) $y = e^{5x}$

(c) $y = e^{-x}$

(d) $y = 3e^{2x}$

(e) $y = 5e^{-4x}$

a) $y = e^x$ $\quad \frac{dy}{dx} = e^x$

$\frac{dy}{dx} = y$

AT $x=3$, $\quad \frac{dy}{dx} = e^3$

$\frac{dy}{dx} = e^3$

FINAL ANSWER IN TERMS OF e

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b) $y = e^{5x}$ $\quad \frac{dy}{dx} = 5e^{5x}$

$\frac{dy}{dx} = ke^{kx}$

AT $x=3$, $\quad \frac{dy}{dx} = 5e^{5\times 3}$

$\frac{dy}{dx} = 5e^{15}$

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c) $y = e^{-x}$ $\quad \frac{dy}{dx} = -e^{-x}$

AT $x = 3$, $\quad \frac{dy}{dx} = -e^{-3}$

$\frac{dy}{dx} = -e^{-3} = -\frac{1}{e^3}$

$\frac{dy}{dx} = -ke^{-kx}$

$k = 1$

NOT ESSENTIAL

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d) $y = 3e^{2x}$ $\quad \frac{dy}{dx} = 6e^{2x}$

AT $x = 3$, $\quad \frac{dy}{dx} = 6e^{2\times 3}$

$\frac{dy}{dx} = 6e^{6}$

$\frac{dy}{dx} = ke^{kx}$

3 IS A CONSTANT
SO IS MULTIPLIED BY k

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e) $y = 5e^{-4x}$ $\quad \frac{dy}{dx} = -20e^{-4x}$

AT $x = 3$, $\quad \frac{dy}{dx} = -20e^{-4\times 3}$

$\frac{dy}{dx} = -20e^{-12}$

$\frac{dy}{dx} = -ke^{-kx}$

5 IS A CONSTANT

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